ACM/IDS 104 APPLIED LINEAR ALGEBRA
PRACTICE PROBLEMS FOR LECTURE 8: SOLUTIONS

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Problem 8A. Lagrange Interpolation

Suppose we want to fit a polynomial $y(t)$ of degree n to the data set $(t_1, y_1), \ldots, (t_{n+1}, y_{n+1})$. Then, as we proved in Lecture 8, the polynomial is given by

$$y(t) = y_1 L_1(t) + \ldots + y_{n+1} L_{n+1}(t), \tag{1}$$

where $L_k(t)$ is the kth *Lagrange polynomial*,

$$L_k(t) = \frac{(t-t_1)\ldots(t-t_{k-1})(t-t_{k+1})\ldots(t-t_{n+1})}{(t_k-t_1)\ldots(t_k-t_{k-1})(t_k-t_{k+1})\ldots(t_k-t_{n+1})}, \tag{2}$$

which is the unique polynomial of degree n such that $L_k(t_k) = 1$ and $L_k(t_i) = 0$ for $i \neq k$. The interpolating polynomial $y(t)$ is often referred to as the *Lagrange interpolating polynomial*.

Lagrange interpolation is used for estimating the value $f(t^*)$ of a function $f(t)$ when the explicit expression for $f(t)$ is unknown, but its values are measured at other points: $y_1 = f(t_1), \ldots, y_{n+1} = f(t_{n+1})$. Lagrange interpolation has applications in image processing, weather forecasting, and financial and astronomical modeling. However, as we will see in Problem 4 of PS4, Lagrange interpolation can exhibit large oscillations between data points (*Runge's phenomenon*) when using high-degree polynomials, and it can be computationally expensive for a large number of data points. In such cases, alternative interpolation methods like cubic splines are preferred.

Consider the exponential function $f(t) = e^t$.

(a) Find the Lagrange interpolating polynomial $y(t)$ which interpolates $f(t)$ at $t_1 = -1, t_2 = 0$, and $t_3 = 1$.
(b) Plot $f(t)$, $y(t)$, and the approximating polynomial $x(t)$ of the same degree as $y(t)$ obtained by the truncated Taylor expansion of $f(t)$ at zero.

Solution:

(a) Since we have three data points, in this problem $n = 2$. The first three Lagrange polynomials are:

$$\begin{aligned} L_1(t) &= \frac{(t-t_2)(t-t_3)}{(t_1-t_2)(t_1-t_3)} = \frac{t(t-1)}{2} = \frac{1}{2}t^2 - \frac{1}{2}t, \\ L_2(t) &= \frac{(t-t_1)(t-t_3)}{(t_2-t_1)(t_2-t_3)} = \frac{(t+1)(t-1)}{-1} = -t^2 + 1, \\ L_3(t) &= \frac{(t-t_1)(t-t_2)}{(t_3-t_1)(t_3-t_2)} = \frac{(t+1)t}{2} = \frac{1}{2}t^2 + \frac{1}{2}t. \end{aligned} \tag{3}$$

The interpolating polynomial is, therefore,

$$\begin{aligned} y(t) &= f(t_1)L_1(t) + f(t_2)L_2(t) + f(t_3)L_3(t) = \frac{e^{-1}}{2}t^2 - \frac{e^{-1}}{2}t - t^2 + 1 + \frac{e}{2}t^2 + \frac{e}{2}t \\ &= \left(\frac{e+e^{-1}}{2} - 1\right)t^2 + \left(\frac{e-e^{-1}}{2}\right)t + 1. \end{aligned} \tag{4}$$

(b) The Taylor series expansion for $f(t)$ at $t = 0$, truncated after the second-order term, is

$$f(t) \approx 1 + t + \frac{1}{2}t^2 =: x(t). \tag{5}$$

Numerically,

$$y(t) \approx 1 + 1.175t + 0.543t^2. \tag{6}$$

Figure 1 shows the comparison between the original function $f(t)$, its Lagrange interpolating polynomial $y(t)$, and its Taylor approximating polynomial $x(t)$.

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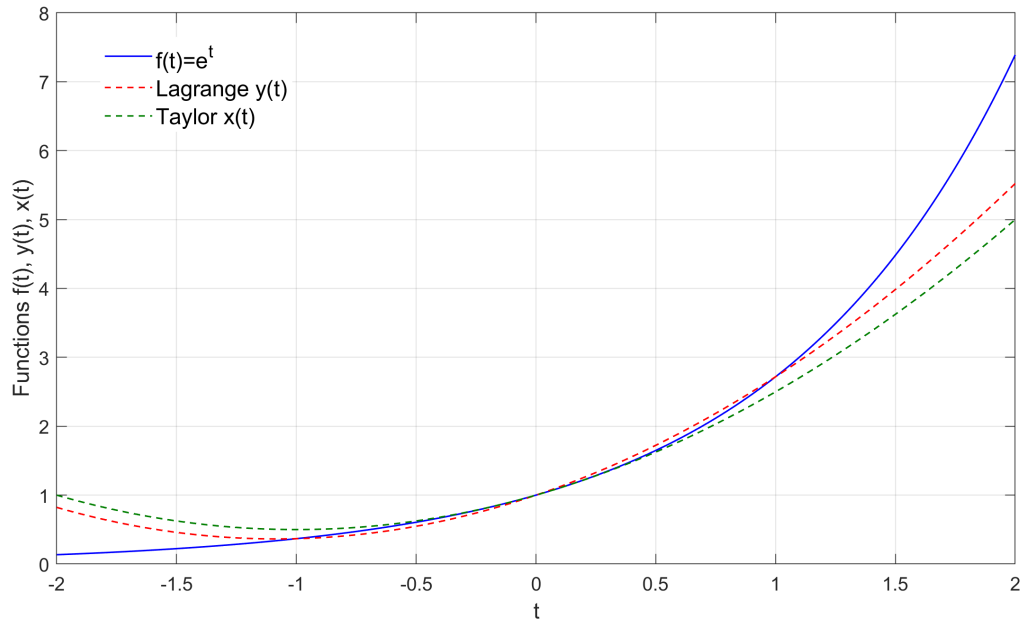

FIGURE 1. The function $f(t) = e^t$, its Lagrange interpolating polynomial $y(t)$, and its Taylor approximating polynomial $x(t)$.

Problem 8B. VANDERMONDE MATRIX

In Lecture 8, we encountered a *Vandermonde matrix*, which naturally arose in the context of the polynomial least squares fitting. The $(n+1) \times (n+1)$ Vandermonde matrix is

$$V = \begin{bmatrix} 1 & t_1 & t_1^2 & \dots & t_1^n \\ 1 & t_2 & t_2^2 & \dots & t_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_{n+1} & t_{n+1}^2 & \dots & t_{n+1}^n \end{bmatrix} \tag{7}$$

The Vandermonde matrix arises in the discrete Fourier transform (in this context it is often called the DFT matrix), in statistics and polynomial regression (where it is often called the design matrix), and in the representation theory of the symmetric group.

Prove that the determinant of the Vandermonde matrix can be written in the following elegant form:

$$\det V = \prod_{1 \le i < j \le n+1} (t_j - t_i). \tag{8}$$

In particular, this formula shows that V is nonsingular if and only if all $t_1, \dots, t_{n+1}$ are distinct.

Solution: The proof is based on the fact that the determinant of a matrix is invariant under the elementary column operation of type I. In other words, if we add to a column a_i a multiple of another column a_j, that is, if we replace a_i with $a_i + \alpha a_j$, then the determinant remains unchanged.

Let's perform the following sequence of elementary column operations on the Vandermonde matrix:

$$a_{n+1} \longrightarrow a_{n+1} - t_1 a_n, \qquad a_n \longrightarrow a_n - t_1 a_{n-1}, \qquad \dots \qquad a_2 \longrightarrow a_2 - t_1 a_1. \tag{9}$$

Then the Vandermonde matrix will be transformed as follows:

$$V(t_1, \dots, t_{n+1}) = \begin{bmatrix} 1 & t_1 & t_1^2 & \dots & t_1^n \\ 1 & t_2 & t_2^2 & \dots & t_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_{n+1} & t_{n+1}^2 & \dots & t_{n+1}^n \end{bmatrix} \longrightarrow$$
$$\begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 1 & t_2 - t_1 & t_2(t_2 - t_1) & t_2^2(t_2 - t_1) & \dots & t_2^{n-1}(t_2 - t_1) \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_{n+1} - t_1 & t_{n+1}(t_{n+1} - t_1) & t_{n+1}^2(t_{n+1} - t_1) & \dots & t_{n+1}^{n-1}(t_{n+1} - t_1) \end{bmatrix} = V' \tag{10}$$

Since the determinant is invariant under this transformation, $\det V = \det V'$. Expanding the determinant of V' along the first row gives

$$\det V(t_1, \ldots, t_{n+1}) = \det \begin{bmatrix} t_2 - t_1 & t_2(t_2 - t_1) & t_2^2(t_2 - t_1) & \ldots & t_2^{n-1}(t_2 - t_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_{n+1} - t_1 & t_{n+1}(t_{n+1} - t_1) & t_{n+1}^2(t_{n+1} - t_1) & \ldots & t_{n+1}^{n-1}(t_{n+1} - t_1) \end{bmatrix} \tag{11}$$

All entries of the ith row of the matrix on the right-hand side have $(t_{i+1} - t_1)$ as a common factor. Therefore,

$$\det V(t_1, \ldots, t_{n+1}) = (t_2 - t_1) \ldots (t_{n+1} - t_1) \det V(t_2, \ldots, t_{n+1}), \tag{12}$$

where

$$V(t_2, \ldots, t_{n+1}) = \begin{bmatrix} 1 & t_2 & t_2^2 & \ldots & t_2^{n-1} \\ 1 & t_3 & t_3^2 & \ldots & t_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_{n+1} & t_{n+1}^2 & \ldots & t_{n+1}^{n-1} \end{bmatrix} \tag{13}$$

is the $n \times n$ Vandermonde matrix based on $t_2, \ldots, t_{n+1}$. Now we can use (12), apply the same argument to $\det V(t_2, \ldots, t_{n+1})$, and keep iterating until we reduce the matrix size to 2×2:

$$\begin{aligned} &\det V(t_1, \ldots, t_{n+1}) = (t_2 - t_1) \ldots (t_{n+1} - t_1) \det V(t_2, \ldots, t_{n+1}) \\ &= (t_2 - t_1) \ldots (t_{n+1} - t_1) \times (t_3 - t_2) \ldots (t_{n+1} - t_2) \det V(t_3, \ldots, t_{n+1}) \\ &= (t_2 - t_1) \ldots (t_{n+1} - t_1) \times (t_3 - t_2) \ldots (t_{n+1} - t_2) \times \ldots \times (t_n - t_{n-1})(t_{n+1} - t_{n-1}) \det \begin{bmatrix} 1 & t_n \\ 1 & t_{n+1} \end{bmatrix} \\ &= (t_2 - t_1) \ldots (t_{n+1} - t_1) \times (t_3 - t_2) \ldots (t_{n+1} - t_2) \times \ldots \times (t_n - t_{n-1})(t_{n+1} - t_{n-1}) \times (t_{n+1} - t_n) \\ &= \prod_{1 \le i < j \le n+1} (t_j - t_i). \end{aligned} \tag{14}$$